

3. This question is about geostationary and geosynchronous satellites.

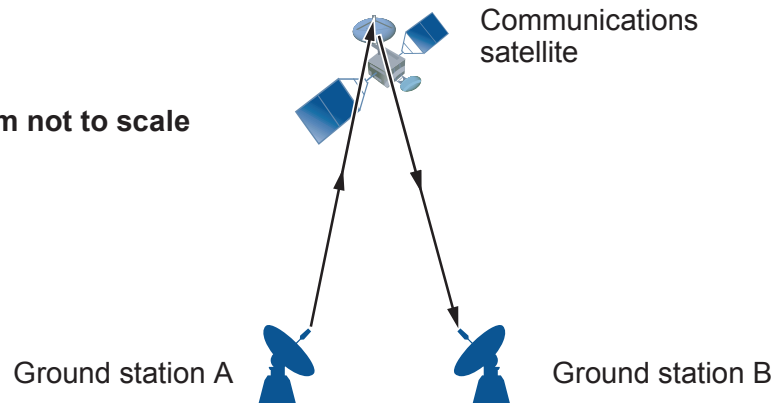
- (a) **Place a tick (✓) or a cross (×) in each box** in the table below.
One box has been completed for you.

[4]

	Geosynchronous satellite	Geostationary satellite
Stays above the same point on Earth at all times	×	
Orbits Earth once in 24 hours		
Orbits above the equator		
Must pass over the North pole and South pole		

- (b) The diagram shows a communications satellite relaying a signal between ground stations A and B.

Diagram not to scale



- (i) Satellite communication uses microwaves of wavelength of 0.08 m.
Use the equation:

$$\text{wave speed} = \text{frequency} \times \text{wavelength}$$

to calculate the frequency of the microwaves and state its unit.
(Speed of light, $c = 3 \times 10^8 \text{ m/s}$)

[4]

Frequency =

Unit =



- (ii) The time delay between transmitting the signal at A and receiving it back on Earth at B is 0.24 s.
Use an equation from page 2 to calculate the distance from the satellite to the ground stations. [3]
(Speed of light, $c = 3 \times 10^8$ m/s)

Distance = m

11

3420UA01
11

